VMM PO Governance — SAP Interface API Reference (v1.4)

For: SAP ABAP / workflow / Basis developers building the SAP side of the PO Governance check. **Contract:** *SAP API Specification v1.4* and `po-governance-sap-api.yaml` (OpenAPI 3.0) in `legacy/python-app/docs/`. This page lists every endpoint with its request and response as the tool answers them today. All examples below were captured from the running interface.

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1. Connection

Item	Staging (SAP QA)	Production
Base URL	`http://192.168.0.175:8178/api/sap/v1`	`https://<vm-dns-name>:8443/api/sap/v1` (company certificate)
SM59 destination	Type G, e.g. `ZPOGOV_API_TEST` — host `192.168.0.175`, port `8178`, path prefix `/api/sap/v1`, SSL off	Type G `ZPOGOV_API` — port `8443`, SSL active (import the company CA chain in STRUST)
Technical user	`sap_ecq` (one per SAP system)	`sap_ecp`
Connection test	`GET /ping` → `200`	same

Authentication

- **HTTP Basic authentication** with a technical user (SM59 → *Logon & Security* → *Basic Authentication*). The tool's administrator creates the user and gives you the password, generated with 32 characters, through a secure channel.
- Wrong or missing credentials → `401 UNAUTHORIZED`, with header `WWW-Authenticate: Basic`.
- After 5 wrong passwords in a row the user is locked for 15 minutes → `401 USER_LOCKED`.
- An optional IP allow-list can be switched on by the tool's administrator. A caller outside the list → `401 IP_NOT_ALLOWED`.

Rules for every call

- UTF-8 JSON, `Content-Type: application/json`. Field names are **snake_case**, exactly as written here.
- Timestamps are ISO 8601 with offset, e.g. `2026-10-22T14:24:58+05:30`. Dates are `YYYY-MM-DD`.
- Quantities are JSON numbers in the base unit (normally `PC`). Amounts are in document currency.
- Article numbers (`article`, `generic_article`) are sent in **SAP internal format with leading zeros**; the tool removes them.
- **OTB is never sent and never checked by the tool**; it stays in SAP.
- Every answer carries an `X-Trace-Id` header (e.g. `T-a1490d`). Quote it when you report a problem.

Error body (every non-200 answer)

```
{ "error_code": "VALIDATION_FAILED", "message": "purchase_order.items[0].order_qty must be > 0",
"trace_id": "T-a19102" }
```

error_code	HTTP	Meaning	What SAP does
`UNAUTHORIZED`	401	Missing / wrong user or password	Fix SM59 credentials
`USER_LOCKED`	401	Too many wrong passwords, or user disabled	Ask the tool admin to unlock

error_code	HTTP	Meaning	What SAP does
`IP_NOT_ALLOWED`	401	Caller IP not in the allow-list (only when switched on)	Ask the tool admin to add the IP
`INVALID_JSON`	400	Body is not valid JSON	Fix the program; do not resend unchanged
`VALIDATION_FAILED`	400	Mandatory field missing (`/decisions`: generic_article, colour, order_qty, delivery_date), wrong value, quantity ≤ 0, bad date (`message` names the field)	Fix the program; do not resend unchanged
`REQUEST_ID_CONFLICT`	409	`/decisions`: same `request_id` already used with a different body	Generate a new `request_id`
`CANNOT_DECIDE`	422	`/decisions`: an item cannot be decided (e.g. article unknown to the tool)	Treat as **NOT_CHECKED**, continue as today
`BATCH_TOO_LARGE`	413	`/events`: more than 500 events or 5 MB	Split the batch and resend
`REQUEST_NOT_FOUND`	404	`/documents`: no answered decision for the `request_id`	Use the `request_id` of a `/decisions` call that answered 200
`DOCUMENT_CONFLICT`	409	`/documents`: another document is already stored for the `request_id`	Send `replace = true` if it is a correction
`DECISION_TIMEOUT`, `DATA_NOT_LOADED`, `DATABASE_UNAVAILABLE`, `SAP_INTERFACE_DISABLED`, `INTERNAL_ERROR`	503	Tool temporarily unable to answer	`/decisions`: **NOT_CHECKED**, continue as today. `/events`: retry

2. Endpoints at a glance

#	Method & path	Called by	Purpose
1	`GET /ping`	SM59 connection test; health job every minute	Connection and credential check
2	`GET /reason-codes`	SAP approver screen	Exception reasons for the dropdown
3	`POST /decisions`	Release workflow background step "PO Governance check" (after OTB + depth check passed)	**Governance verdict per PO item** (synchronous, SAP waits max 3 s)
4	`POST /events`	bgRFC queue from BAdIs (after commit)	PO, goods receipt / return and governance-outcome events (asynchronous)
5	`POST /documents`	After the PO is really created / received in SAP	**Final document number (MRN) against the `request_id`** of the decision — feeds the "PO generated" report
6	`POST /mra-status`	Every time an MRA changes in SAP during the day (created, discarded, rejected, self-rejected …)	**MRA status per line**; the morning MRA file in the tool's Data Setup is merged into the same lines

3. `GET /ping`

Connection test. If 3 health checks fail in a row, SAP switches the governance step off and runs the previous process.

Request: no body; Basic authentication only.

Response `200`:

```
{
  "status": "ok",
  "server_time": "2026-10-22T14:24:58+05:30",
  "api_version": "1.0.0",
  "environment": "TEST",
  "source_system_recognised": "ECQ"
}
```

Field	Type	Meaning
`status`	string	Always `ok` when the answer is 200
`server_time`	date-time	Tool server time
`api_version`	string	Interface version
`environment`	`TEST` | `PRODUCTION`	Which copy of the tool answered
`source_system_recognised`	string	SAP system ID registered for the technical user (`ECQ`, `ECP`) — check it matches `SY-SYSID`

Other answers: `401` (credentials), `503` (tool down).

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4. `GET /reason-codes`

Exception reasons the approver can choose when asking for an exception (send the chosen `code` back in a `GOV_OUTCOME` event). Maintained by Planning in the tool; read it when the screen opens or cache it daily.

Request: no body.

Response `200`:

```
{
  "version": "2026-10-22",
  "reason_codes": [
    { "code": "NEWL", "text": "New launch - no sales history", "active": true },
    { "code": "FEST", "text": "Festive / event build-up", "active": true },
    { "code": "VMOQ", "text": "Vendor minimum order quantity", "active": true },
    { "code": "PRCA", "text": "Price advantage / one-time buy", "active": true },
    { "code": "REPL", "text": "Store replenishment gap", "active": true },
    { "code": "OTHR", "text": "Other (note mandatory)", "active": true }
  ]
}
```

Field	Type	Meaning
`version`	string	Date of the last change — refresh your cache when it changes
`reason_codes[].code`	string, max 4	Code to show / store
`reason_codes[].text`	string, max 60	Text to show
`reason_codes[].active`	boolean	`false` = do not offer any more

The list above is the proposal from the specification and is **still to be confirmed by the business**.

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5. `POST /decisions` — the verdict API

Called by the workflow background step **after SAP's OTB check and depth plan check have passed**. SAP sends the PO; the tool answers with a verdict per item.

Timing and fallback contract

- The tool answers in **~20–60 ms** (measured; 50 parallel POs all under 1 s). SAP waits **at most 3 seconds**.
- On timeout, network error or **any non-200 answer**, SAP marks the PO `NOT_CHECKED` and continues the workflow exactly as today. **The tool can never stop SAP.**
- **Mode:** send `SHADOW` for now. The tool decides and logs, but SAP must **not** route on the verdict. `LIVE` = SAP routes on `route_to`. Control it in SAP, e.g. switch `ZPOGOV_ACTIVE` per division.
- **Idempotency:** `request_id` identifies one check (e.g. work-item ID + PO + version). Repeating the same `request_id` with the same body returns the **same answer** (same `decision_id`). The same `request_id` with a different body → `409 REQUEST_ID_CONFLICT`. If SAP leaves `request_id` out, the tool builds one (`AUTO-yyyyMMdd-<body hash>`), so the same body sent again on the same day also returns the same answer.

Request

Business rule: only four fields per item are mandatory — `generic_article`, `colour`, `order_qty` and `delivery_date` (Generic Material + Colour + Delivery Month + Proposed Qty). Everything else is optional. The smallest valid request:

```
{
  "purchase_order": {
    "items": [
      { "generic_article": "000000001101000101", "colour": "WHT", "order_qty": 50000,
"delivery_date": "2026-11-15" }
    ]
  }
}
```

The full request, with every optional field SAP may send for audit and reporting:

```
{
  "request_id": "WI000004711-4500099001-V1",
  "mode": "SHADOW",
  "source_system": "ECQ",
  "client": "300",
  "otb_check": "PASSED",
  "depth_check": "PASSED",
  "previous_decision_id": null,
  "requested_at": "2026-10-21T11:50:00+05:30",
  "purchase_order": {
    "po_number": "4500099001",
    "doc_type": "NB",
    "vendor": "0000201345",
    "purch_org": "1000",
    "purch_group": "B07",
    "buyer_user": "BUYER07",
    "buyer_email": "ravi.k@vishalwholesale.co.in",
    "division": "MENS",
    "currency": "INR",
    "items": [
      {
        "po_item": 10,
        "article": "000000001101000101",
        "generic_article": "000000001101000101",
        "colour": "WHT",
        "merch_category": "MC1021",
        "season": "AW26",
        "site": "DC01",
        "order_qty": 50000,
        "uom": "PC",
        "net_value": 418800.00,
        "delivery_date": "2026-11-15"
      }
    ]
  }
}
```


Field	Type	Req.	SAP source / meaning	When left out
`purchase_order.items[]`	array, min 1	**yes**	One entry per PO item	—
`items[].generic_article`	string	**yes**	`MARA-SATNR` (generic article), leading zeros kept	—
`items[].colour`	string	**yes**	Colour characteristic of the variant: Colour Master **code** preferred (description also accepted)	—
`items[].order_qty`	number, > 0	**yes**	`EKPO-MENGE` in base unit (proposed quantity)	—
`items[].delivery_date`	date	**yes**	First schedule line `EKET-EINDT` — the tool uses its month	—
`request_id`	string, max 80	no	Unique per check, e.g. work-item ID + PO + version	`AUTO-yyyyMMdd-<body hash>`
`mode`	`SHADOW` | `LIVE`	no	See above	`SHADOW`
`source_system`	string	no	`SY-SYSID`	System of the technical user (`ECQ` / `ECP`)
`client`	string	no	`SY-MANDT`	—
`otb_check`	`PASSED`	no	Assertion that OTB passed	`PASSED`
`depth_check`	`PASSED` | `BYPASSED`	no	`BYPASSED` only for a quantity reduced under a previous REDUCE decision	`PASSED`

Field	Type	Req.	SAP source / meaning	When left out
`previous_decision_id`	string | null	no	On resubmission after a REDUCE: the `decision_id` it was reduced under	—
`requested_at`	date-time	no	When SAP sent the check	—
`purchase_order.po_number`	string	no	`EKKO-EBELN` (recommended: shown in the audit log and matched with `GOV_OUTCOME`)	—
`purchase_order.doc_type`	string	no	`EKKO-BSART`	—
`purchase_order.vendor`	string	no	`EKKO-LIFNR`	—
`purchase_order.purch_org`	string	no	`EKKO-EKORG`	—
`purchase_order.purch_group`	string	no	`EKKO-EKGRP`	—
`purchase_order.buyer_user`	string	no	Buyer's SAP user	—
`purchase_order.buyer_email`	string | null	no	Buyer's e-mail	—
`purchase_order.division`	string	no	Division	—
`purchase_order.currency`	string	no	`EKKO-WAERS`	—
`items[].po_item`	integer	no	`EKPO-EBELP` (recommended, so SAP can match the answer to its item)	10, 20, 30 … in the order sent
`items[].article`	string	no	`EKPO-MATNR` (variant); used only when the generic article is not known to the tool	—
`items[].merch_category`	string	no	`EKPO-MATKL`	—
`items[].season`	string	no	Season	—
`items[].site`	string	no	`EKPO-WERKS` (informational; the tool works at company level)	—
`items[].uom`	string	no	`EKPO-MEINS`	—
`items[].net_value`	number	no	Item net value	—

An optional field that **is** sent must still be valid (e.g. `mode` must be `SHADOW` or `LIVE`, `po_item` must not repeat), otherwise the answer is `400 VALIDATION_FAILED`. The answer always echoes `request_id`, `mode` and every `po_item`, including the ones the tool filled in.

How the tool reads an item:

- **Article:** the generic article with its leading zeros removed (`000000001101000101` → `1101000101`), looked up in the tool's Article Master / Universe Master (then `article`, if sent).
- **Colour:** matched by Colour Master code first, then by description.
- **Month:** the delivery month is taken from `delivery_date`. A date in the past counts as the current month.
- **Size variants:** items of one PO with the same generic article, colour and delivery month are decided **together** on their summed quantity. A REDUCE `max_qty` is split over them in proportion to their quantities.

Response `200`


```
{
  "decision_id": "D-20261022-000002",
  "request_id": "WI-REDUCE-0001",
  "mode": "SHADOW",
  "overall_verdict": "REDUCE",
  "verdict_code": "R",
  "route_to": "BUYER",
  "bm_head_required": false,
  "summary_text": "Item 10: reduce to 15,928 - Stock + pending + this PO cover 265 days, above the
79-day norm; max 15,928 pcs",
  "rules_version": "R-P1",
  "data_delayed": false,
  "data_as_of": { "bw_snapshot": "2026-10-22T14:24:36+05:30", "events_lag_seconds": 0 },
  "decided_at": "2026-10-22T14:24:58+05:30",
  "view_url": "http://192.168.0.175:8177/verdict/R_LVm2VaqeZN94W6EOgf44SuA27lJ9VQr5rUyQx5hb0",
  "view_url_expires_at": "2027-01-20T14:24:58+05:30",
  "items": [
    {
      "po_item": 10,
      "verdict": "REDUCE",
      "verdict_code": "R",
      "max_qty": 15928,
      "reason_code": "COVER_ABOVE_NORM",
      "reason_text": "Stock + pending + this PO cover 265 days, above the 79-day norm; max 15,928
pcs",
      "rule_id": "COVER-01",
      "metrics": { "plan_qty": 6303, "stock_qty": 2707, "open_po_qty": 2703, "headroom_qty": 15928,
"cover_days": 265 }
    }
  ]
}
```


Field	Type	Meaning / what SAP does with it
`decision_id`	string	e.g. `D-20261022-000002`. **Store it on the PO / workflow container** for audit and send it back in `GOV_OUTCOME` events
`request_id`	string	Echo of the request
`mode`	`SHADOW` | `LIVE`	Echo of the request
`overall_verdict`	`ALLOW` | `REDUCE` | `WATCH` | `HOLD`	Most severe item verdict (HOLD > WATCH > REDUCE > ALLOW)
`verdict_code`	`A` | `R` | `W` | `H`	One-character code for SAP domain values
`route_to`	`NONE` | `BUYER` | `CATHEAD` | `BMHEAD`	Where the workflow goes in LIVE mode (ignored in SHADOW). `NONE` = continue normal approvals
`bm_head_required`	boolean	Escalation to the B&M head (business rule pending — currently always `false`)
`summary_text`	string, max 255	Short text for the work item / PO header note
`rules_version`	string	Version of the planner parameters used
`data_delayed`	boolean	`true` = the tool's event feed is behind the agreed limit → **SAP continues the PO as today (NOT_CHECKED)**
`data_as_of.bw_snapshot`	date-time	Time of the stock data the decision used

Field	Type	Meaning / what SAP does with it
`data_as_of.events_lag_seconds`	integer	How far behind the event feed is
`decided_at`	date-time	When the tool decided
`view_url`	string (URL)	**Link to the verdict page** on the PO Governance portal. Show it to the buyer / approver (work item, PO note, e-mail): clicking it opens the verdict with its items and reasons — read-only, **no sign-in needed**. The token in the link is random and cannot be guessed; anyone who has the link can open it, so keep it inside the company. Omitted if the link is switched off
`view_url_expires_at`	date-time	Until when the link works (90 days by default)

The verdict page (`view_url`) shows the complete verdict: what happens next for this verdict, the decision details (decision / request ID, decided at, route, mode, buyer, division, SAP system, rules version, stock data date, final document), every item with its reason, rule and the figures behind it (plan, stock, open PO, headroom, cover days or sell-through), totals, and a print button. A decision answered before the link existed gets its `view_url` the first time SAP repeats the same `request_id` (the answer is otherwise unchanged and stays the same on every later repeat). | `items[].po_item` | integer | PO item | | `items[].verdict` / `verdict_code` | as above | Item verdict | | `items[].max_qty` | number | null | **Maximum allowed quantity when the verdict is REDUCE**; `null` otherwise | | `items[].reason_code` | string | Machine-readable reason (table below) | | `items[].reason_text` | string, max 132 | Text for the buyer / approver | | `items[].rule_id` | string | Rule that produced the verdict | | `items[].metrics` | object | Figures behind the verdict. Single Concept: `plan_qty`, `stock_qty`, `open_po_qty`, `headroom_qty`, `cover_days`. Multi Concept: `sell_through_pct`. Fields that do not apply are omitted |

`cover_days` is the cover from the first day of the month after delivery. It can be `0`: that means current stock plus pending POs is sold out before the delivery window starts.

Verdicts — what each one means for SAP

Verdict	Code	`route_to`	SAP action (LIVE mode)
`ALLOW`	A	`NONE`	Continue normal approvals
`REDUCE`	R	`BUYER`	Buyer reduces the item to `max_qty` or less. Depth plan bypass: store `decision_id` + `max_qty` on the item; if the quantity is reduced to `max_qty` or less, resubmit with `depth_check = BYPASSED` and `previous_decision_id`
`WATCH`	W	`CATHEAD`	Category head approves or rejects (exception with reason code)
`HOLD`	H	`CATHEAD`	Do not release as is; category head decides. **A blocked article (slow seller) is sent as `HOLD` with `reason_code = BLOCK`**

Reason codes per verdict

Verdict	reason_code	rule_id	Meaning
ALLOW	`COVER_WITHIN_NORM`	COVER-00	Stock + pending + this PO within the cover norm
ALLOW	`NO_PLAN_DATA`	PLAN-00	No sales plan for the bucket — allowed
ALLOW	`PEER_Q1` / `PEER_Q2`	PEER-Q1 / PEER-Q2	Multi Concept: ranks in the top half of its peer group
ALLOW	`PEER_Q3`	PEER-Q3-OK	Q3 but at / near the peer average (border zone)
ALLOW	`BOBS_SIBLINGS_OK`	BOBS-01	New colour of a selling style; sibling colours sell well
ALLOW	`LOW_HISTORY`	WOF-00	Too little selling history to judge
ALLOW	`NO_PEER_POOL`	PEER-00	No comparable styles
ALLOW	`NEW_ARTICLE`	NEW-00	New article (not in the tool's history)
REDUCE	`COVER_ABOVE_NORM`	COVER-01	Cover above the norm — reduce to `max_qty`
WATCH	`MIX_BELOW_MIN`	MIX-01	Colour share below the minimum mix (broken / cut size likely)

Verdict	reason_code	rule_id	Meaning
WATCH	`PEER_Q3`	PEER-Q3-GAP	Q3, beyond the border zone
WATCH	`PEER_Q4`	PEER-Q4-WATCH	Q4 but at / near the peer average
WATCH	`BOBS_SIBLING_LOW`	BOBS-02	A sibling colour sells poorly
HOLD	`COVER_NO_HEADROOM`	COVER-02	Stock + pending already above the norm — no headroom (max 0)
HOLD	`BLOCK`	BLOCK-Q4	Q4, well below the peer average: confirmed slow seller
HOLD	`BLOCK`	BLOCK-ABS	Absolute block: sell-thru below 1 %/day after 4+ weeks on floor

The tool may later send `BLOCK` (code `B`) as its own verdict instead of `HOLD` + `BLOCK`, once SAP has added that domain value. This is switched in the tool's settings and will be agreed with you first. Until then, handle `reason_code = BLOCK` under `HOLD`.

More examples (captured)

ALLOW, with the colour sent as its description (`"colour": "wht desc"`):


```
{ "decision_id": "D-20261022-000001", "request_id": "WI-ALLOW-0001", "mode": "SHADOW",
  "overall_verdict": "ALLOW", "verdict_code": "A", "route_to": "NONE", "bm_head_required": false,
  "summary_text": "All 1 item(s) within the norms", "rules_version": "R-P1", "data_delayed": false,
  "data_as_of": { "bw_snapshot": "2026-10-22T14:24:36+05:30", "events_lag_seconds": 0 },
  "decided_at": "2026-10-22T14:24:58+05:30",
  "items": [ { "po_item": 10, "verdict": "ALLOW", "verdict_code": "A", "max_qty": null,
               "reason_code": "COVER_WITHIN_NORM", "reason_text": "Cover 0 days within the 79-day
norm", "rule_id": "COVER-00",
               "metrics": { "plan_qty": 6303, "stock_qty": 2707, "open_po_qty": 2703,
"headroom_qty": 15928, "cover_days": 0 } } ] }
```


HOLD (no headroom):


```
{ "...": "...", "overall_verdict": "HOLD", "verdict_code": "H", "route_to": "CATHEAD",
  "summary_text": "Item 10: hold - Stock + pending already above the 79-day cover norm; no headroom
(max 0)",
  "items": [ { "po_item": 10, "verdict": "HOLD", "verdict_code": "H", "max_qty": null,
               "reason_code": "COVER_NO_HEADROOM", "reason_text": "Stock + pending already above the
79-day cover norm; no headroom (max 0)",
               "rule_id": "COVER-02",
               "metrics": { "plan_qty": 7108, "stock_qty": 25604, "open_po_qty": 3698,
"headroom_qty": 0, "cover_days": 101 } } ] }
```


WATCH (colour mix):


```
{ "...": "...", "mode": "LIVE", "overall_verdict": "WATCH", "verdict_code": "W", "route_to":
"CATHEAD",
  "summary_text": "Item 10: watch - Colour mix 0.0% below the 2% minimum of its MC-MRP",
  "items": [ { "po_item": 10, "verdict": "WATCH", "verdict_code": "W", "max_qty": null,
               "reason_code": "MIX_BELOW_MIN", "reason_text": "Colour mix 0.0% below the 2% minimum
of its MC-MRP",
               "rule_id": "MIX-01", "metrics": { "stock_qty": 874, "open_po_qty": 0 } } ] }
```


Block (sent as HOLD + BLOCK):

```
{ "...": "...", "overall_verdict": "HOLD", "verdict_code": "H", "route_to": "CATHEAD",
  "summary_text": "Item 10: hold - Ranks Q4, sell-thru 64.4% below the peer average: confirmed slow
seller",
  "items": [ { "po_item": 10, "verdict": "HOLD", "verdict_code": "H", "max_qty": null,
               "reason_code": "BLOCK", "reason_text": "Ranks Q4, sell-thru 64.4% below the peer
average: confirmed slow seller",
               "rule_id": "BLOCK-Q4", "metrics": { "sell_through_pct": 0.97 } } ] }
```


Several items: two size variants of one generic article and colour (decided together, `max_qty` split 9,557 + 6,371 = 15,928) plus a Multi Concept item:


```
{ "decision_id": "D-20261022-000006", "request_id": "WI-MULTI-0001", "mode": "SHADOW",
  "overall_verdict": "REDUCE", "verdict_code": "R", "route_to": "BUYER", "bm_head_required": false,
  "summary_text": "Item 10: reduce to 9,557 - …; Item 20: reduce to 6,371 - …",
  "items": [
    { "po_item": 10, "verdict": "REDUCE", "verdict_code": "R", "max_qty": 9557, "reason_code":
"COVER_ABOVE_NORM",
      "reason_text": "Stock + pending + this PO cover 265 days, above the 79-day norm; max 15,928
pcs for 2 items together (50,000 pcs ordered)",
      "rule_id": "COVER-01", "metrics": { "plan_qty": 6303, "stock_qty": 2707, "open_po_qty": 2703,
"headroom_qty": 15928, "cover_days": 265 } },
    { "po_item": 20, "verdict": "REDUCE", "verdict_code": "R", "max_qty": 6371, "reason_code":
"COVER_ABOVE_NORM",
      "reason_text": "Stock + pending + this PO cover 265 days, above the 79-day norm; max 15,928
pcs for 2 items together (50,000 pcs ordered)",
      "rule_id": "COVER-01", "metrics": { "plan_qty": 6303, "stock_qty": 2707, "open_po_qty": 2703,
"headroom_qty": 15928, "cover_days": 265 } },
    { "po_item": 30, "verdict": "ALLOW", "verdict_code": "A", "max_qty": null, "reason_code":
"PEER_Q1",
      "reason_text": "Ranks Q1 in its peer group (sell-thru 5.2%/day): healthy seller", "rule_id":
"PEER-Q1",
      "metrics": { "sell_through_pct": 5.2 } } ] }
```


Error answers of `/decisions` (captured)

Case	HTTP	Body
Article unknown to the tool (whole PO not decided)	422	`{"error_code":"CANNOT_DECIDE","message":"Item 10: Article 9999999999 is unknown to the tool (not in Article / Universe Master, F2 or BSR)","trace_id":"T-a1490d"}`
Same `request_id`, different body	409	`{"error_code":"REQUEST_ID_CONFLICT","message":"request_id WI-ALLOW-0001 was already used with a different body (decision D-20261022-000001) - generate a new request_id","trace_id":"T-a4d243"}`
`colour` missing	400	`{"error_code":"VALIDATION_FAILED","message":"purchase_order.items[0].colour is required","trace_id":"…"}`
`order_qty` = 0	400	`{"error_code":"VALIDATION_FAILED","message":"purchase_order.items[0].order_qty must be > 0","trace_id":"T-a19102"}`
Broken JSON	400	`{"error_code":"INVALID_JSON","message":"Invalid JSON: …","trace_id":"T-1e66c8"}`
No / wrong credentials	401	`{"error_code":"UNAUTHORIZED","message":"Missing or wrong credentials","trace_id":"T-809985"}`

If **any** item cannot be decided, the whole request is answered `422`. The message names the item(s), and SAP treats the PO as NOT_CHECKED. Every non-200 answer from `/decisions` = **NOT_CHECKED, continue as today**.

---

6. `POST /events`

Called from a **bgRFC unit after commit**. One call carries **1–500 events** in the order they occurred (ascending `seq`).

Delivery contract

- `200` = every event in the batch is safely stored (as accepted, duplicate or parked). **Remove the unit from the queue.**
- Any other status, or no answer within **10 seconds → retry the same batch**. Resending is always safe: events are de-duplicated by `msg_id`.
- An event that fails validation is **parked** on the tool side (listed in `parked`, alerted to IT) and **never blocks the queue**.
- **PO events carry the full current state of the PO** (header + all items), not only what changed.
- Scope: vendor POs and goods movements against vendor POs (receipts 101/102, returns 122/161). Stock transfers (STO) are not sent.

Request

```
{
  "source_system": "ECQ",
  "client": "300",
  "queue": "ZPOGOV_EVENTS",
  "sent_at": "2026-10-21T11:42:08+05:30",
  "events": [ { "...one or more events, see below...": "" } ]
}
```

Field	Type	Req.	Meaning
`source_system`	string	**yes**	`SY-SYSID`
`client`	string	**yes**	`SY-MANDT`
`queue`	string	no	bgRFC queue name, e.g. `ZPOGOV_EVENTS`
`sent_at`	date-time	no	When the unit was sent
`events`	array, 1–500	**yes**	The events

Every event:

Field	Type	Req.	Meaning
`msg_id`	string, max 80	**yes**	Globally unique, built by SAP (system + client + object + timestamp / counter). Used for de-duplication
`seq`	integer (int64)	**yes**	Ascending number per queue, assigned when the event is written
`event_type`	enum	**yes**	One of the 11 types below
`occurred_at`	date-time	**yes**	When the document was saved in SAP
`sap_user`	string	no	`SY-UNAME`
`purchase_order` / `goods_movement` / `governance_outcome`	object	**one of them**	Filled according to `event_type`

event_type	Payload
`PO_CREATED`, `PO_CHANGED`, `PO_RELEASED`, `PO_RELEASE_REVOKED`, `PO_CLOSED`, `PO_DELETED`	`purchase_order` (full current state)
`GR_POSTED`, `GR_REVERSED`, `RETURN_POSTED`, `RETURN_REVERSED`	`goods_movement` (one material-document line)

event_type	Payload
`GOV_OUTCOME`	`governance_outcome` (what happened after a verdict)

`purchase_order` (PO events)

```
{
  "msg_id": "ECQ300-PO-4500098765-20261021101502-01", "seq": 45098, "event_type": "PO_RELEASED",
  "occurred_at": "2026-10-21T10:15:02+05:30", "sap_user": "BUYER07",
  "purchase_order": {
    "po_number": "4500098765", "doc_type": "NB", "vendor": "0000201345", "purch_org": "1000",
"purch_group": "B07",
    "created_on": "2026-10-18", "created_by": "BUYER07", "release_status": "RELEASED",
"release_indicator": "R",
    "currency": "INR", "deleted": false,
    "items": [
      { "po_item": 10, "article": "000000001101000101", "generic_article": "000000001101000101",
"colour": "WHT",
        "merch_category": "1101", "site": "DC01", "order_qty": 600, "received_qty": 0,
"returned_qty": 0, "open_qty": 600,
        "uom": "PC", "net_price": 349.00, "price_unit": 1, "delivery_date": "2026-11-10",
        "deletion_flag": false, "delivery_completed": false }
    ]
  }
}
```

Field	Req.	SAP source
`po_number`	**yes**	`EKKO-EBELN`
`doc_type`, `vendor`, `purch_org`, `purch_group`	no	`EKKO-BSART`, `LIFNR`, `EKORG`, `EKGRP`
`created_on`, `created_by`	no	`EKKO-AEDAT / BEDAT`, `ERNAM`
`release_status`	no	`NOT_RELEASED` | `IN_RELEASE` | `RELEASED` | `REFUSED` (from `FRGKE` / `FRGZU`)
`release_indicator`, `currency`	no	`EKKO-FRGKE`, `WAERS`
`deleted`	no	`true` if the whole PO is deleted / cancelled
`items[]`	**yes**	All items
`items[].po_item`, `article`, `order_qty`, `uom`	**yes**	`EKPO-EBELP`, `MATNR`, `MENGE`, `MEINS`
`items[].generic_article`, `colour`	no (send them)	`MARA-SATNR`, colour characteristic
`items[].merch_category`, `site`	no	`EKPO-MATKL`, `WERKS`
`items[].received_qty`, `returned_qty`, `open_qty`	no	From `EKBE`; `open_qty = order − received + returned`, 0 if completed / deleted
`items[].net_price`, `price_unit`	no	`EKPO-NETPR`, `PEINH`
`items[].delivery_date`	no	First schedule line `EKET-EINDT`
`items[].deletion_flag`, `delivery_completed`	no	`EKPO-LOEKZ`, `ELIKZ`

`goods_movement` (GR / return events)

```
{
  "msg_id": "ECQ300-GR-5000123456-2026-0001", "seq": 45121, "event_type": "GR_POSTED",
  "occurred_at": "2026-10-21T11:42:07+05:30", "sap_user": "DCUSER01",
  "goods_movement": {
    "material_doc": "5000123456", "doc_year": 2026, "doc_item": 1, "movement_type": "101",
"debit_credit": "S",
    "posting_date": "2026-10-21", "article": "000000001101000101", "generic_article":
"000000001101000101",
    "colour": "WHT", "site": "DC01", "storage_location": "0001", "quantity": 240, "uom": "PC",
    "po_number": "4500098765", "po_item": 10, "reversal_of": null
  }
}
```

Reversal (`GR_REVERSED`, movement 102) and return (`RETURN_POSTED`, movement 122):

```
{ "msg_id": "ECQ300-GR-5000123457-2026-0001", "seq": 45130, "event_type": "GR_REVERSED",
"occurred_at": "2026-10-21T11:50:00+05:30",
  "sap_user": "DCUSER01",
  "goods_movement": { "material_doc": "5000123457", "doc_year": 2026, "doc_item": 1,
"movement_type": "102", "debit_credit": "H",
                      "posting_date": "2026-10-21", "article": "000000001101000101", "quantity": 40,
"uom": "PC",
                      "po_number": "4500098765", "po_item": 10, "reversal_of": "5000123456" } }

{ "msg_id": "ECQ300-RT-5000123500-2026-0001", "seq": 45140, "event_type": "RETURN_POSTED",
"occurred_at": "2026-10-21T12:00:00+05:30",
  "goods_movement": { "material_doc": "5000123500", "doc_year": 2026, "doc_item": 1,
"movement_type": "122", "debit_credit": "H",
                      "article": "000000001101000101", "quantity": 10, "uom": "PC", "po_number":
"4500098765", "po_item": 10 } }
```

Field	Req.	SAP source
`material_doc`, `doc_year`, `doc_item`	**yes**	`MBLNR`, `MJAHR`, `ZEILE`
`movement_type`	**yes**	`BWART` (101, 102, 122, 161 …)
`article`, `quantity`, `uom`	**yes**	`MATNR`, `MENGE` (always positive, base unit), unit
`debit_credit`	no	`S` | `H` (`SHKZG`) — the direction
`posting_date`	no	`BUDAT`
`generic_article`, `colour`	no (send them)	Generic article, colour
`site`, `storage_location`	no	`WERKS`, `LGORT`
`po_number`, `po_item`	no	`EBELN`, `EBELP` if the movement refers to a PO
`reversal_of`	no	Original material document (`SMBLN`) for a reversal

`governance_outcome` (`GOV_OUTCOME`)

What happened after a verdict (buyer accepted, exception approved / rejected …). It feeds the audit trail.

```
{
  "msg_id": "ECQ300-OUT-D-20261021-000001-01", "seq": 45188, "event_type": "GOV_OUTCOME",
  "occurred_at": "2026-10-21T15:05:12+05:30", "sap_user": "CATHEAD03",
  "governance_outcome": {
    "decision_id": "D-20261021-000001", "po_number": "4500099001", "po_item": 10,
    "outcome": "EXCEPTION_APPROVED", "by_user": "CATHEAD03", "by_role": "CATHEAD",
    "at": "2026-10-21T15:05:12+05:30", "new_qty": null, "reason_code": "FEST", "note": "Diwali
build-up for DC01"
  }
}
```

Field	Req.	Meaning
`decision_id`	**yes**	From the `/decisions` answer
`po_number`	**yes**	PO
`po_item`	no	Item (null = whole PO)
`outcome`	**yes**	`BUYER_ACCEPTED_REDUCTION` | `EXCEPTION_REQUESTED` | `EXCEPTION_APPROVED` | `EXCEPTION_REJECTED` | `PO_RELEASED` | `PO_WITHDRAWN` | `NOT_CHECKED_TIMEOUT`
`by_user`	**yes**	SAP user
`by_role`	no	`BUYER` | `CATHEAD` | `BMHEAD` | `SYSTEM`
`at`	**yes**	date-time
`new_qty`	no	New quantity (e.g. after accepting a reduction)
`reason_code`	no	Code from `GET /reason-codes`
`note`	no	Max 500 characters (mandatory for `OTHR` by business rule)

Response `200`

```
{
  "received_at": "2026-10-22T14:24:59+05:30",
  "accepted": [
    "ECQ300-PO-4500098765-20261021101502-01",
    "ECQ300-GR-5000123456-2026-0001",
    "ECQ300-GR-5000123457-2026-0001",
    "ECQ300-RT-5000123500-2026-0001",
    "ECQ300-OUT-D-20261021-000001-01"
  ],
  "duplicates": [],
  "parked": [
    { "msg_id": "ECQ300-GR-5000123999-2026-0001", "error_code": "VALIDATION_FAILED", "message":
"goods_movement.article is required" },
    { "msg_id": "ECQ300-XX-1", "error_code": "VALIDATION_FAILED",
      "message": "event_type PO_SHIPPED is not one of PO_CREATED, PO_CHANGED, PO_RELEASED,
PO_RELEASE_REVOKED, PO_CLOSED, PO_DELETED, GR_POSTED, GR_REVERSED, RETURN_POSTED, RETURN_REVERSED,
GOV_OUTCOME" }
  ]
}
```

Field	Meaning
`received_at`	When the tool stored the batch
`accepted`	`msg_id`s stored now
`duplicates`	`msg_id`s already received earlier (not stored again) — sending the same batch twice gives all of them here
`parked`	Events that failed validation, with the reason; IT is alerted. **Still a 200 — do not resend these**

Other answers: `400` for a malformed envelope (e.g. `{"error_code":"VALIDATION_FAILED","message":"client is required",…}`), `401` for credentials, `413` for more than 500 events or 5 MB (split), `503` for a temporary problem (retry).

---

7. `POST /documents` — final document (MRN) of a decided PO

Called when the PO is **really created in SAP** (e.g. after release / goods receipt, whatever document the business names "final", normally the MRN). It saves the document number against the `request_id` of the `/decisions` call, so the tool can report which checked POs were actually generated — not only which ones were asked about.

Rules

- One current document per `request_id`. Sending **the same** `document_no` again is safe → `status: UNCHANGED`.
- A **different** `document_no` for the same `request_id` → `409 DOCUMENT_CONFLICT`, unless `replace = true` (a correction: the old number is kept in the history) → `status: REPLACED`.
- The `request_id` must belong to a `/decisions` call that answered **200**; otherwise `404 REQUEST_NOT_FOUND`.
- Same Basic authentication as the other endpoints. Retry on timeout / 503 — resending is always safe.

Request

```
{
  "request_id": "WI000004711-4500099001-V1",
  "document_no": "5000123456",
  "document_type": "MRN",
  "po_number": "4500099001",
  "document_date": "2026-10-26",
  "replace": false,
  "note": ""
}
```

Field	Type	Req.	Meaning
`request_id`	string, max 80	**yes**	The `request_id` sent to `/decisions` (when SAP did not send one, the `request_id` the tool returned)
`document_no`	string, max 40	**yes**	Final document number (MRN)
`document_type`	string, max 20	no	Default `MRN`
`po_number`	string, max 20	no	`EKKO-EBELN`; default: the PO number of the decision
`document_date`	date	no	`YYYY-MM-DD`
`replace`	boolean	no	`true` = correct a document already stored for this `request_id`
`note`	string, max 500	no	Free text

Response `200` (captured)

```
{ "request_id": "WI-REDUCE-0001", "decision_id": "D-20261026-000001", "document_no": "5000123456",
  "document_type": "MRN", "status": "STORED", "received_at": "2026-10-26T14:55:03+05:30" }
```

Field	Meaning
`status`	`STORED` (first document) | `UNCHANGED` (same document again) | `REPLACED` (`replace = true`)
`decision_id`	The decision the document belongs to
`received_at`	When the tool stored it

Error answers (captured)

Case	HTTP	Body
Another document already stored	409	`{"error_code":"DOCUMENT_CONFLICT","message":"request_id WI-REDUCE-0001 already has document 5000123456 - send replace = true to correct it","trace_id":"T-4834f8"}`

Case	HTTP	Body
`request_id` never decided	404	`{"error_code":"REQUEST_NOT_FOUND","message":"No answered decision for request_id WI-NOPE - post the document with the request_id of a /decisions call that answered 200","trace_id":"T-9deafd"}`
`document_no` missing	400	`{"error_code":"VALIDATION_FAILED","message":"document_no is required","trace_id":"T-00125f"}`

8. `POST /mra-status` — MRA status changes during the day

The tool keeps the current status of every MRA line; it will be used in the verdicts later on (open MRA quantity).

- **Each morning** the planner uploads the MRA report in the tool (📂 Data Setup → MRA Status). The file is **merged** into the tool's MRA list: its lines are added or updated (qty, status and every filled column); lines that are not in the file stay as they are, with the changes SAP sent.
- **During the whole day** SAP sends every change of an MRA to this endpoint, as soon as it happens: a new MRA, and an MRA that is discarded, rejected or self-rejected (or any other status change, or a quantity change).

Rules

- **Required per line:** **`mra_no`**, **`qty`**, **`status`**. All other fields are optional. A field that is not sent keeps the value the tool already has (from the morning file or an earlier message).
- 1–500 lines per call, **applied in the order sent**, in one transaction. If **any** line is wrong, the whole call is refused (`400`) and nothing is stored, so correct it and send the call again.
- **Line of an MRA:** `mra_no` + `generic_article` + `colour`. For an MRA with **one** line, `mra_no` alone is enough. For an MRA with **several** lines (colours / articles), send `generic_article` and `colour`; otherwise that line is not changed (`AMBIGUOUS`).
- **`changed_at`** (recommended): when the change happened in SAP. A message with an **older** `changed_at` than the one the tool has is ignored (`STALE`), so messages that arrive late or are retried never undo a newer change.
- **Safe to resend**: the same message again gives `UNCHANGED`.
- `status` values (spelling is not case-sensitive; spaces and hyphens are allowed):

`status`	Meaning	Also accepted
`ALLOW`	MRA created after an ALLOW verdict	`Allow`, `Allowed`
`WATCHLIST`	MRA created after a WATCH verdict	`Watchlist`, `Watch List`, `Watch`
`CANCELLED`	MRA cancelled	`Cancelled`, `Canceled`, `Cancel`
`DISCARDED`	MRA discarded	`Discard`, `Discarded`
`REJECTED`	MRA rejected by an approver	`Reject`, `Rejected`
`SELF_REJECTED`	MRA rejected by the buyer	`Self Reject`, `Self-Rejected`

More values can be added in the tool's settings (`SapInterface:MraStatuses`) without a change for SAP.

Request

Minimal (status change of an MRA with one line):

```
{ "mras": [ { "mra_no": "0000317231", "qty": 1200, "status": "SELF_REJECTED", "changed_at": "2026-11-04T10:15:00+05:30" } ] }
```

Full (a new MRA, every field):

```
{ "mras": [
  { "mra_no": "0000317250", "generic_article": "1101000150", "colour": "YLW", "qty": 75, "status":
"ALLOW",
    "changed_at": "2026-11-04T10:20:00+05:30", "request_id": "WI000004711-4500099001-V1",
    "generic_art_desc": "MENS KURTA", "vendor_code": "0000201345", "vendor_name": "ABC TEXTILES",
"brand_type": "NON-PL",
    "mc_code": "121030002", "mc_name": "L_EW_MM_KURTA", "mrp": 799, "planning_flag": "NON-BOGO",
"fit": "REGULAR",
    "design": "SOLID", "style": "KURTA", "fabric": "COTTON", "product_material": "COTTON",
"po_value": 59925.00,
    "delivery_date": "2026-12-31", "approval_level": "Level 2", "remark": "" } ] }
```


Field	Type	Req.	Meaning
`mras`	array, 1–500	**yes**	The changes, in the order they happened
`mras[].mra_no`	string, max 20	**yes**	MRA number (keep the leading zeros)
`mras[].qty`	number ≥ 0	**yes**	MRA quantity (pieces) after the change
`mras[].status`	string	**yes**	See the table above
`mras[].generic_article`	string, max 40	for MRAs with several lines	Generic article of the line
`mras[].colour`	string, max 40	for MRAs with several lines	Colour of the line
`mras[].changed_at`	timestamp	recommended	When the change happened in SAP (ISO 8601 with offset)
`mras[].request_id`	string, max 80	no	The tool's `request_id` of the check that allowed the MRA
`mras[].generic_art_desc`	string, max 200	no	Article description
`mras[].vendor_code`, `vendor_name`	string, max 20 / 200	no	Vendor
`mras[].brand_type`	string, max 40	no	e.g. `NON-PL`
`mras[].mc_code`, `mc_name`	string, max 20 / 100	no	Merchandise category
`mras[].mrp`	number	no	MRP
`mras[].planning_flag`	string, max 20	no	e.g. `BOGO` / `NON-BOGO`
`mras[].fit`, `design`, `style`, `fabric`, `product_material`	string, max 100	no	Attributes (Multi Concept)
`mras[].po_value`	number	no	Value
`mras[].delivery_date`	date	no	`YYYY-MM-DD`
`mras[].approval_level`	string, max 30	no	Where the MRA is waiting, e.g. `Level 2`
`mras[].remark`	string, max 500	no	Free text

Response `200` (captured)


```
{ "batch_id": "MRA-20261104-160840-5d1e", "received": 3, "created": 1, "updated": 2, "unchanged": 0,
"stale": 0, "ambiguous": 0,
  "results": [
    { "index": 0, "mra_no": "0000317231", "generic_article": null, "colour": null, "result":
"UPDATED",
      "previous_status": "ALLOW", "status": "SELF_REJECTED", "qty": 1200 },
    { "index": 1, "mra_no": "0000317232", "generic_article": null, "colour": null, "result":
"UPDATED",
      "previous_status": "WATCHLIST", "status": "DISCARDED", "qty": 300 },
    { "index": 2, "mra_no": "0000317250", "generic_article": "1101000150", "colour": "YLW",
"result": "CREATED",
      "previous_status": null, "status": "ALLOW", "qty": 75 } ],
  "received_at": "2026-11-04T16:08:40+05:30" }
```


`result`	Meaning	SAP action
`CREATED`	New MRA line stored	–
`UPDATED`	Status / quantity / fields changed	–
`UNCHANGED`	Same as already stored (a resend)	–
`STALE`	`changed_at` older than the stored change: ignored	– (a newer change is already there)
`AMBIGUOUS`	The MRA has several lines and no `generic_article` / `colour` was sent: **not stored**	Send the line again with `generic_article` and `colour`

Error answers (captured)

Case	HTTP	Body
A required field missing	400	`{"error_code":"VALIDATION_FAILED","message":"mras[0].qty is required","trace_id":"T-c1f11c"}`
Unknown status	400	`{"error_code":"VALIDATION_FAILED","message":"mras[0].status 'APPROVED' is not one of ALLOW, WATCHLIST, CANCELLED, DISCARDED, REJECTED, SELF_REJECTED","trace_id":"…"}`
More than 500 lines	413	`{"error_code":"BATCH_TOO_LARGE","message":"501 MRA lines in one batch - at most 500","trace_id":"…"}`

`401` for credentials, `503` for a temporary problem (retry; resending is safe).

---

9. Status of the tool side

Item	Status
`ping`, `reason-codes`, `decisions`, `events`, `documents`, `mra-status`	**Available for testing** on staging
MRA status in the verdicts	MRA lines are stored now (morning file + `mra-status`); using them in the verdicts (open MRA quantity) is the next step
Verdict link (`view_url`)	Available; the portal page `/verdict/<token>` opens without sign-in
Decision logic	Same rules as the tool's New Request screen (Single Concept, Multi Concept, Watch List, Max Permissible)
Data used for decisions	Uploaded BW / SAP extracts (stock, pending PO, sales, plan, BSR) + the day's approved requests
Events → live open-PO / stock position	**In development**: events are received and stored now, and will feed the live position. Until then decisions use the uploaded files, so run **SHADOW** mode

Item	Status
Separate `BLOCK` verdict, B&M head limit (`bm_head_required`), final reason-code list	To be confirmed with the business
Multi Concept `max_qty` (REDUCE for Multi Concept items)	Rules pending from the business

Suggested test sequence for the SAP team:

1. SM59 connection test (`GET /ping`).
2. `GET /reason-codes`.
3. `POST /decisions` in SHADOW mode for a few real POs (ALLOW, REDUCE, WATCH, HOLD cases), then the same call again to check you get the same `decision_id`.
4. Timeout / NOT_CHECKED handling (e.g. a wrong password gives 401 → must continue as today).
5. `POST /events` for a PO save, a goods receipt and a GOV_OUTCOME, then the same batch again to see `duplicates`.
6. Open a `view_url` from a decision answer in a browser.
7. `POST /documents` with the MRN of a PO that was created, then the same call again (`UNCHANGED`).
8. `POST /mra-status`: a new MRA (`ALLOW`), then the same MRA `SELF_REJECTED`, then the first message again (`STALE`).

The tool's administrator sees every call — request, response, timing and trace ID — on 🔌 **SAP Interface** in the portal, so results can be checked together.